On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to zero grade and is subject to harsh discipline issues.

Zeliha Nur GABUK
150719050

---

1) consider the matrix $A = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$ Some of the elimination matrix given below

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}, \underline{\begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix}}, \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & 6 \\ 6 & 6 & -1/2 & 0 \end{bmatrix}$$

a) Find the unknown entries α, γ, β, k and m.

yukarıda verilen ikinci elimination matrixe göre $m \rightarrow ⓶$ $\boxed{-2}$

ve $-3\alpha = -8$ $\quad \alpha = \boxed{\frac{8}{3}}$

$$\begin{bmatrix} 1 & 8/3 & -\gamma & 8/3 \\ 8 & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & 64/9 \\ 1 & 14 & -6 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 8/3 & -\gamma & 8/3 \\ 0 & -49/3 & -\gamma & -61/3 \\ 0 & & & \\ 0 & & & \end{bmatrix}$$

2) A geometry in 'k' dimensional space is given by $x=5y$, where 'x' and 'y' represents the axis. The intersection with another geometry gives a volume.

a) give the minimum dimensions of the space with your valuable comments.

minimum dimension of the space is ⑤ çünkü iki tane geometry'nin kesişimi her zaman bulunduğu spaceden daha az olur. Bu durumda 2 tane pivot olacağı için minimum 5D space içerisinde olmalıdır.

b) propose an expression for the unknown geometry to have a volume intersection.

$$x+2y+3z+t+4k=0$$
$$x-5y=0$$

$$\begin{bmatrix}1&2&3&1&4\\1&-5&0&0&0\end{bmatrix}\begin{bmatrix}x\\y\\z\\t\\k\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$$

$$\begin{array}{ccccc}\begin{bmatrix}1&2&3&1&4\\0&7&3&1&4\end{bmatrix}\\ \downarrow\ \ \downarrow\ \ \downarrow\ \ \downarrow\ \ \downarrow \\ P\ \ P\ \ F\ \ F\ \ F\end{array}$$

Volume intersectio

Çünkü 3-free var

3 tane basis alacak buradan ve 3 boyutlu bir kesişim olmuş olacak

c)
$$x+4y-z=6$$
$$x-5y=0$$

$$\begin{bmatrix}1&4&-1\\1&-5&0\end{bmatrix}\begin{bmatrix}x\\y\\z\end{bmatrix}=\begin{bmatrix}6\\0\end{bmatrix}$$

$$\left[\begin{array}{ccc|c}1&4&-1&6\\0&9&-1&6\end{array}\right]\longrightarrow\left[\begin{array}{ccc|c}1&4&-1&6\\0&1&-1/9&2/3\end{array}\right]\longrightarrow\left[\begin{array}{ccc|c}1&0&-5/9&10/3\\0&1&-1/9&2/3\\P&P&F&\end{array}\right]$$

$$x_{special}=\begin{bmatrix}5/9\\1/9\\1\end{bmatrix}\alpha \qquad X_{complete}=\begin{bmatrix}5/9\\1/9\\1\end{bmatrix}\alpha+\begin{bmatrix}10/3\\2/3\\0\end{bmatrix}$$

$$X_{particular}=\begin{bmatrix}10/3\\2/3\\0\end{bmatrix}$$

Shift edildiği için vector space değil

1D inside R^3

3) Suppose $\underline{u}$ is a unit vector in $\mathbb{R}^n$. This problem is about the n by n symmetric matrix. $H = I - 2uu^T$

a) Show directly that $H^2 = I$

$$U = \begin{bmatrix} \ \\ \ \end{bmatrix}_{n\times 1} \quad U^T = \begin{bmatrix} \ & \ \end{bmatrix}_{1\times n} \quad U.U^T = \begin{bmatrix} \ & \ \\ \ & \ \end{bmatrix}_{n\times n}$$ symetric matrix

$$H^2 = (I-2UU^T)(I-2UU^T) = \underset{\overset{\downarrow}{I}}{(I^2)} - \underline{4UU^T} + (4UU^T.UU^T) -$$

$$\underbrace{4UU^T = 4UU^TUU^T}$$

Günkü U unit vector ve bu nedenle UU^T çarpımı symmetric matrix.

$$I - \underbrace{4UU^T + 4(UU^T)^2}_{0} = \boxed{I}$$

b) Show that H is an orthonormal matrix.

Orthonormal matrix → orthonormal vectörlerden oluşan matrixe denir.

Orthonormal vector ise → Orthogonal ve unit vectorlere denir.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & 1 \end{bmatrix}_{n\times n} - 2\underbrace{\begin{bmatrix} \ & \ \\ \ & \ \end{bmatrix}_{n\times n}}_{\text{bu matrix orthonormal matrix dir.}}$$

I matrixten orthonormal matrixin 2 katını çıkardırsak yine orthonormal matrix elde ederiz.

c) one eigenvector of H is U itself. Find the corresponding eigen valu

$$\boxed{1}$$

D) is H diagonalizable?

Evet çünkü independent ve Symmetric matrix. Bu yüzden diagonalizable

4) Two tanks are connected, first thank has $v(0)=10$ liters liquid and the second tank has $w(0)=40$ liters of water. The liquid between the thanks are proportional to the differnce $v-w$. ④

a)

$$\left.\begin{aligned} \frac{dv}{dt} &= -v+w \\ \frac{dw}{dt} &= -w+v \end{aligned}\right\} \Rightarrow \frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \underbrace{\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}}_{A}\begin{bmatrix} v \\ w \end{bmatrix}$$

b) Determine percentages of the mixture in tank 1 at time 1 sec.

$$\det(A-\lambda I) = \begin{vmatrix} -1-\lambda & 1 \\ 1 & -1-\lambda \end{vmatrix} = 0 \Rightarrow (-1-\lambda)^2 - 1 = 0$$

$$\lambda^2 + 2\lambda = 0$$

$$(\lambda+2)\lambda = 0$$

$$\lambda_1 = 0 \quad \lambda_2 = -2$$

$$\lambda_1 = 0 \Rightarrow \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\lambda = -2 \Rightarrow \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$U(t) = S_1 e^{\lambda_1 t} C_1 + S_2 e^{\lambda_2 t} C_2$$

$$U(t) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{0 \cdot t} \cdot C_1 + \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-2t} C_2$$

$$U(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} C_1 + \begin{bmatrix} 1 \\ -1 \end{bmatrix} C_2 = \begin{bmatrix} 10 \\ 40 \end{bmatrix}$$

$$C_1 + C_2 = 10 \qquad 2C_1 = 50$$

$$C_1 - C_2 = 40 \qquad C_1 = 25$$

$$C_2 = -15$$

$$U(1) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cancelto{1}{e^{0 \cdot 1}} (25) + \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-2 \cdot 1} \cdot (-15) \Rightarrow \begin{bmatrix} 25 \\ 25 \end{bmatrix} + \begin{bmatrix} -15/e^2 \\ 15/e^2 \end{bmatrix} = \begin{bmatrix} 25 - 15/e^2 \\ 25 + 15/e^2 \end{bmatrix}$$

mixture in the tank 1

$25 - 15/e^2$

c) Find how much liters of liquids are exist in each tank in steady state?

$$U(\infty) = \begin{bmatrix}1\\1\end{bmatrix} \underbrace{e^{0\cdot t\to\infty}}_{1} \cdot 25 + \begin{bmatrix}1\\-1\end{bmatrix} \underbrace{e^{-2\cdot t\to\infty}}_{0} (-15)$$

$$U(\infty) = \begin{bmatrix}25\\25\end{bmatrix}$$

25 liters liquids in tank 1

25 liters liquids in tank 2